

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

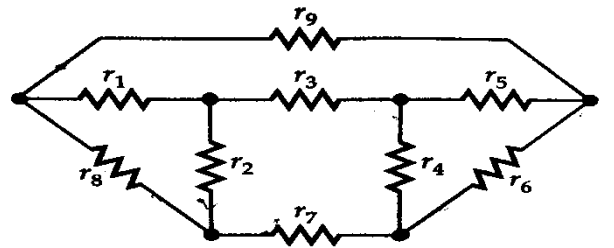
(Level/CO) Marks

Q. 1 Solve Any Two of the following.

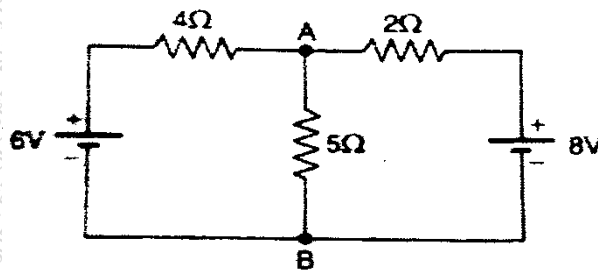
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|---|-------------------------|---|
| A) Explain the difference between short circuit (SC) and open circuit (OC) with an example. | Understand | 6 |
| B) Explain the following terms with suitable example
1. Linear and non linear elements
2. Unilateral and bilateral elements | Understand | 6 |
| C) Explain two types of energy source. Distinguish between ideal and non-ideal sources. | Understand/
Analysis | 6 |

Q.2 Solve Any Two of the following.

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|---|-------------------|---|
| A) State and Explain KVL and KCL with example | Knowledge | 6 |
| B) Following figure represents a resistive network. Draw its graph. Select a suitable tree and obtain the tie-set matrix. | Comprehen
sion | 6 |



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| C) State the Thevenin's theorem and find current through branch AB using Thevenin's theorem. Refer following figure. | Comprehen
sion /
Evaluation | 6 |
|--|-----------------------------------|---|

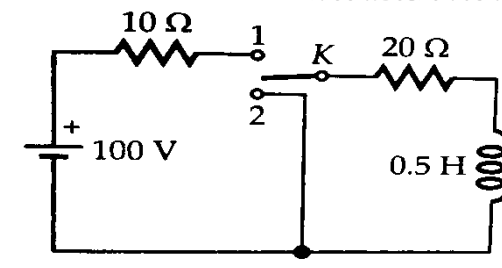


Q. 3 Solve Any Two of the following.

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| A) Explain first order and second order RC and RL circuit. | Understand | 6 |
| B) Explain various properties of a capacitor. | Understand | 6 |
| C) An inductor with initial current I_0 is connected to a resistor of R ohms at $t = 0$. Derive the expression for the current through inductor and voltage across inductor at any time $t > 0$. | Evaluation | 6 |

Q.4 Solve Any Two of the following.

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| A) Derive the symmetry and reciprocity condition of z-parameters. | Evaluation | 6 |
| B) Explain Z-Parameters in terms of Y-parameter & h-parameters. | Understand | 6 |
| C) In following figure the switch K is kept first at position 1 and steady state condition is reached. At $t=0$, the switch is moved to position 2. Find the current in both the cases. | Evaluation | 6 |



Q. 5 Solve Any Two of the following.

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| A) Derive the expression for resonant frequency f_r of a series resonant circuit. | Synthesis | 6 |
| B) Derive the expression for Q factor of parallel resonating circuit. | Synthesis | 6 |
| C) Design a constant K LPF with $f_c = 1\text{KHz}$ and $R_0 = 600\Omega$ At what frequency α will be 10 Db?
To calculate (i) L, C (filter Elements)
(ii) Frequency at which $\alpha = 10\text{dB}$ | Synthesis | 6 |

*** End ***